One Image, Ten Generations: The Hidden Cost of AI Convenience

Conclusion

When "generating more and then selecting" becomes a habit,
the computations carried out by the AI system
far exceed the requirements of the task itself.
We no longer merely select images;
instead, we unconsciously increase the scale of the computation.
A simple act of choice,
triggered an overburdened
infrastructure system.

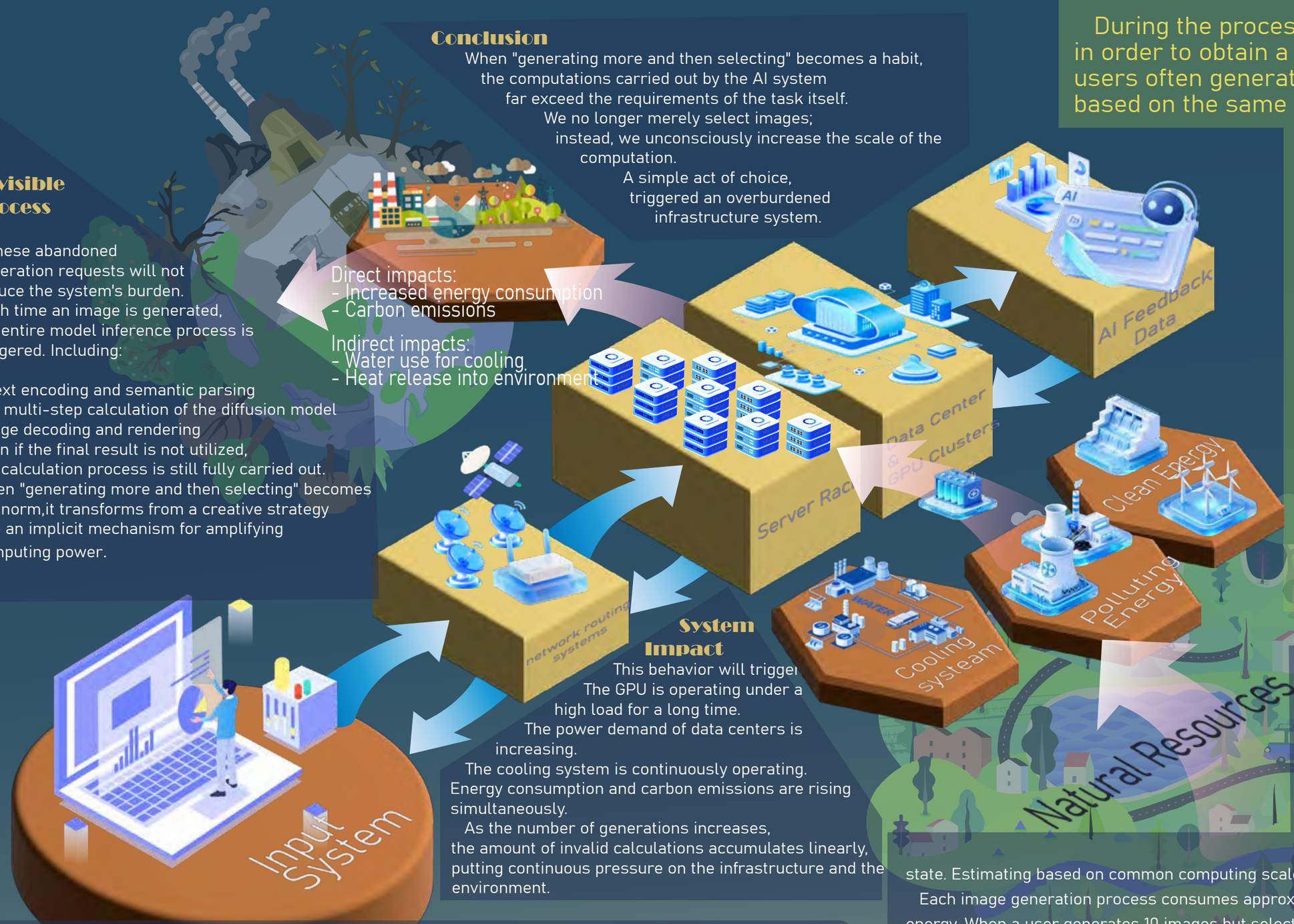
Invisible Process

These abandoned generation requests will not reduce the system's burden.
Each time an image is generated, the entire model inference process is triggered. Including:

Text encoding and semantic parsing
The multi-step calculation of the diffusion model
Image decoding and rendering
Even if the final result is not utilized,
the calculation process is still fully carried out.
When "generating more and then selecting" becomes the norm,it transforms from a creative strategy into an implicit mechanism for amplifying computing power.

Direct impacts:
- Increased energy consumption
- Carbon emissions

Indirect impacts:
- Water use for cooling
- Heat release into environment



System Impact

This behavior will trigger
The GPU is operating under a high load for a long time.
The power demand of data centers is increasing.
The cooling system is continuously operating.
Energy consumption and carbon emissions are rising simultaneously.
As the number of generations increases,
the amount of invalid calculations accumulates linearly,
putting continuous pressure on the infrastructure and the environment.

People's behavioral habits

During the process of AI image generation, in order to obtain a more satisfactory result, users often generate multiple versions based on the same prompt,
and finally select only one of them as the final outcome.
Users generate multiple versions of an image for a single prompt, selecting only one final result. This behavior is common because users seek better visual outcomes through variation and comparison. AI systems enable fast and effortless generation, making repeated attempts feel easy and low-cost.

Energy Consumption Issue

AI image generation relies on the high-intensity parallel computing power of GPUs.
Each time an image is generated, the system will remain in a high-power consumption operating state. Estimating based on common computing scales:
Each image generation process consumes approximately 150 to 300 joules of electrical energy. When a user generates 10 images but selects only 1 of them:
Approximately 90% of the computing and energy consumption is wasted.
In other words:
With every "choice", there is an energy consumption that is several times greater than the outcome itself.

Users generate multiple versions of an image for a single prompt, selecting only one final result.

This behavior is:
- Iterative
- Exploratory
- Convenience-driven
- Excessive